



Chapter 1-2



Question1

() 1. From the view of computer architecture, the machine attributes that assembler programmer can see are _____.

- ☐ A. the functions to be accomplished by computer software.
- ☐ B. the organization of all computer hardware.
- ☐ C. the hardware abstraction to be used when programming.
- ☐ D. the hardware implementation of the various computer components



Question2

- ()2 . Which of the following descriptions about “Computer architecture” is true?
- A. Computer architecture refers only to instruction set architecture design.
 - B. Computer architecture means the implementation of a machine, which has two components: organization and hardware.
 - C. Computer architecture design doesn't need to provide support to compilers.
 - D. Computer architecture is intended to cover all three aspects of computer design ---- instruction set architecture, organization and hardware.



Computer Design Engineering life cycle





Topics in Chapter

1.1 Introduction

1.2 Classes of computers

1.3 Defining computer architecture and What's the task of computer design?

1.4 Trends in Technology

1.5 Trends in power in Integrated circuits

1.6 Trends in Cost

1.7 Dependability

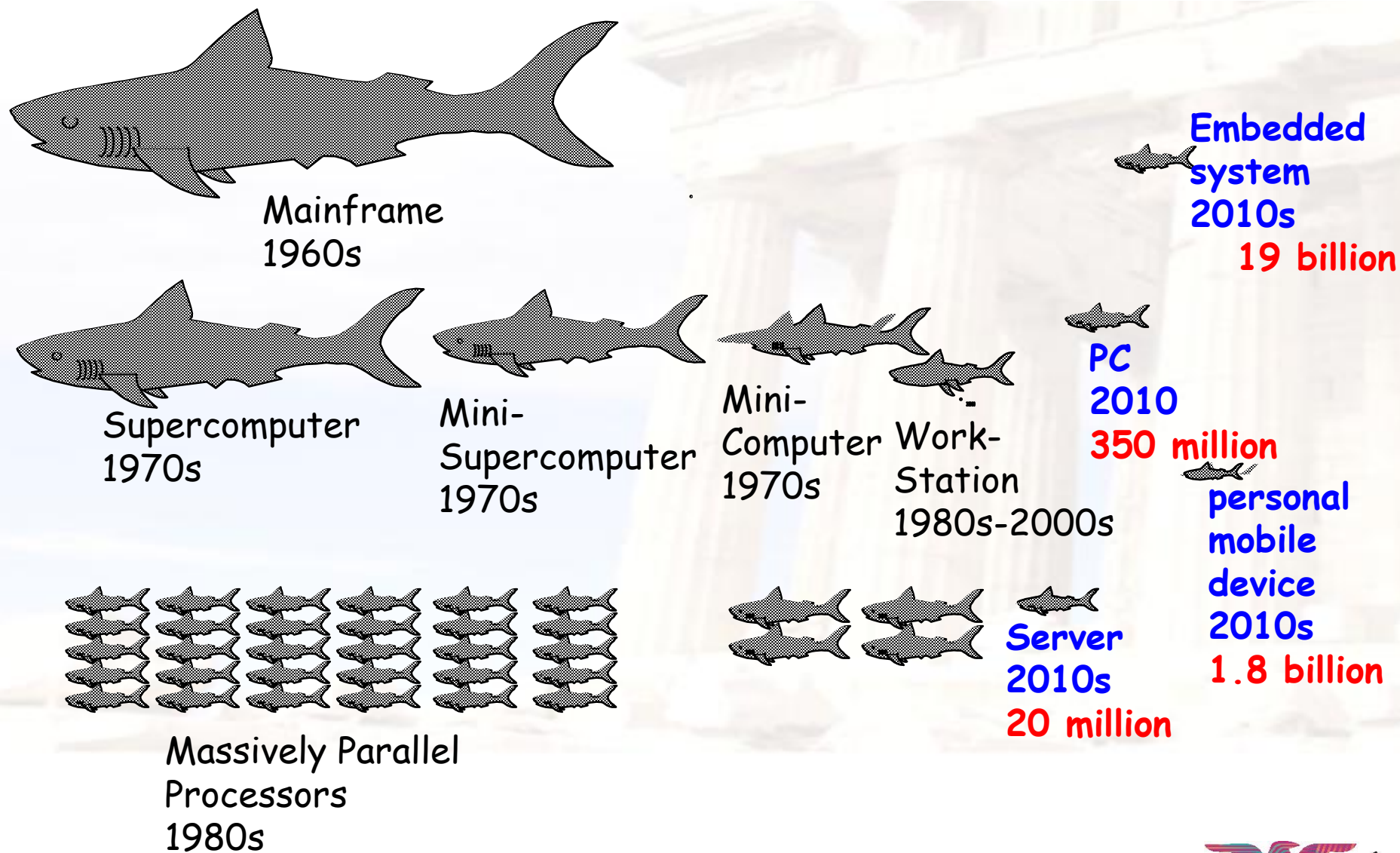
1.8 Measuring, Reporting and summarizing Perf.

1.9 Quantitative Principles of computer Design

1.10 Putting it altogether



Classification-market





5 computing markets

Feature	PMD	Desktop	Server	Clusters	Embedded
Price of system	\$100-\$1000	\$300-\$2500	\$5000-\$10,000,000	\$100000-\$200,000,000	\$10-\$100,000
Price of microprocessor module	\$10-\$100	\$50-\$500 per proc.	\$200-\$2,000 per proc.	\$50-\$250	\$0.01-\$100 per proc.
Critical system design issues	Cost, energy, Media perf. responsiveness	Price-perf., Energy, Graphics perf.	Throughput, availability, Scalability, energy	Price-perf., Throughput, energy proportionality	Price, energy, application-specific perf.



Personal Mobile Devices

❑ Wireless devices with multimedia user interfaces

❑ Application

- Web-based, media-oriented
- Responsiveness, predictability

❑ Requirement:

- Cost, energy efficiency (battery, no-fan)
- Real-time performance
- Minimize memory → flash memory





Desktop Computing

- ❑ The first, and still the largest market in dollar terms, is desktop computing.
(laptop > 50% market of desktop)
- ❑ Requirement:
 - Optimized price-performance
 - highest-perf. & cost-reduced Microprocessor appear first in desktop systems
- ❑ New challenges:
 - Web-centric, interactive application
 - How to evaluate performance ?



Servers

- ❑ The role of servers to provide larger scale and more reliable file and computing services grew.
- ❑ Requirement:
 - First, **availability** is critical.
 - A second key feature of server systems is an emphasis on **scalability**. → **memory, storage** and **I/O bandwidth** is crucial.
 - Lastly, servers are designed for **efficient throughput**.



Cost with downtime

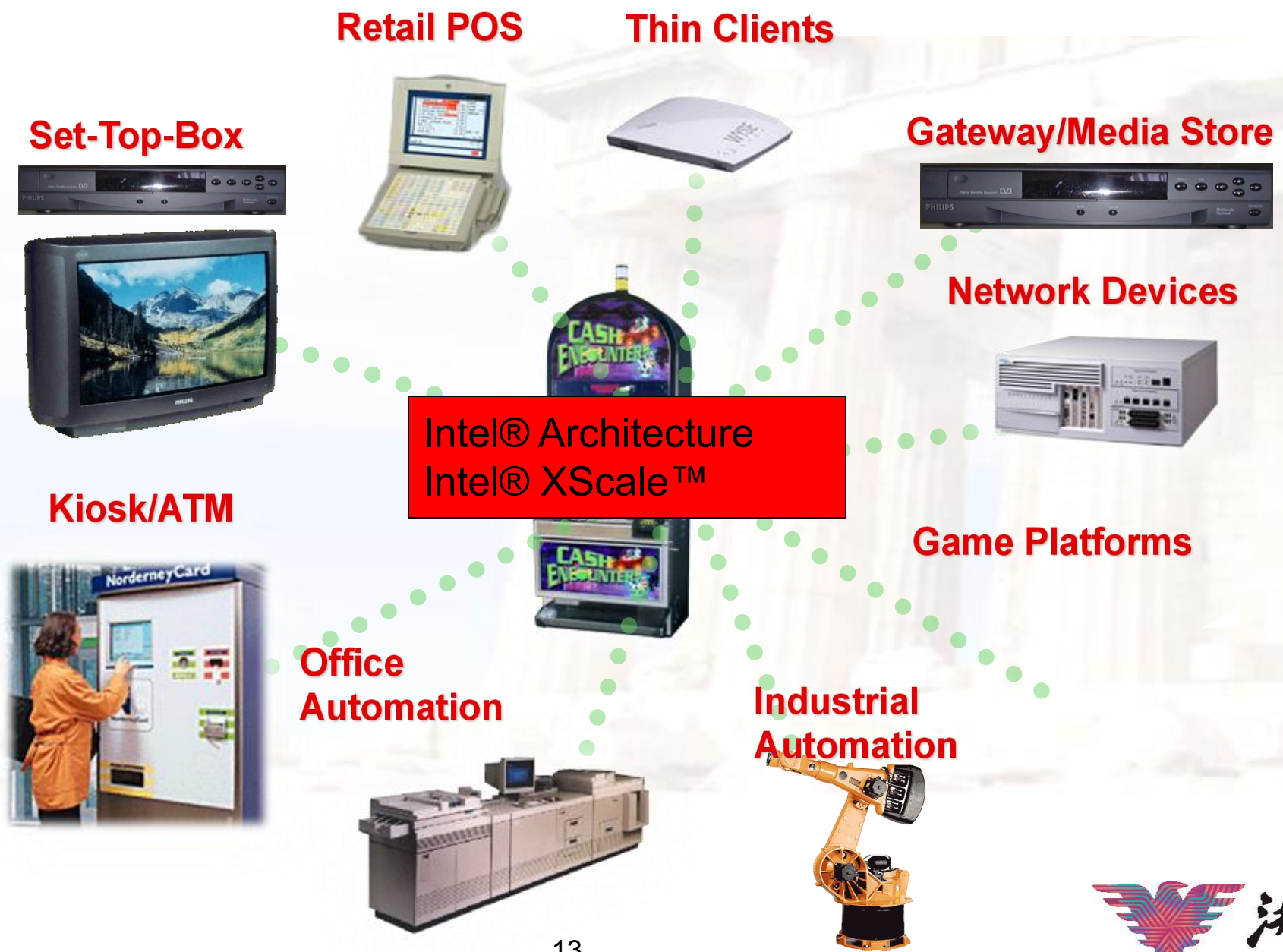
Application	Cost of downtime per hour	1% 87.6 hrs/yr	0.5% 43.8 hrs/yr	0.1% 8.8 hrs/yr
Brokerage operations	\$6,450,000	\$565,000,000	\$283,000,000	\$56,500,000
Credit card authorization	\$2,600,000	\$228,000,000	\$114,000,000	\$22,800,000
Package shipping services	\$150,000	\$13,000,000	\$6,600,000	\$1,300,000
Home shopping channel	\$113,000	\$9,900,000	\$4,900,000	\$1,000,000
Catalog sales center	\$90,000	\$7,900,000	\$3,900,000	\$800,000
Airline reservation center	\$89,000	\$7,900,000	\$3,900,000	\$800,000
Cellular service activation	\$41,000	\$3,600,000	\$1,800,000	\$400,000
Online network fees	\$25,000	\$2,200,000	\$1,100,000	\$200,000
ATM service fees	\$14,000	\$1,200,000	\$600,000	\$100,000





Cluster/Warehouse-Scale Computers

- ❑ SaaS (Search, social networking, video sharing, multiplayer games, online-shopping)
- ❑ WSC—tens of thousands of servers act as one
- ❑ Requirements:
 - Price-performance, Power (10% saves \$7M)
 - Availability is critical
 - Scalability (Note the difference with server)
 - (Note the difference with supercomputer)





Embedded Computers

- ❑ The fastest growing portion of the computer market with widest spread of processing power and cost.
- ❑ Requirements
 - Real time performance (soft & hard)
 - Strict resource constraints
 - limited memory size, lower power consumption,...
- ❑ The use of processor cores together with application-specific circuitry.
 - DSP, mobile computing, mobile phone, Digital TV



Other classification

- ❑ Quantum computer vs Chemical computer,
- ❑ Scalar processor vs Vector processor
- ❑ Non-Uniform Memory Access (NUMA)/ UMA
- ❑ **Register machine vs Stack machine vs Accumulator machine**
- ❑ Harvard architecture vs Von Neumann architecture
/Non Von Neumann architecture （类脑芯片：达尔文2）
- ❑ **RISC vs. CISC**
- ❑ Cellular architecture



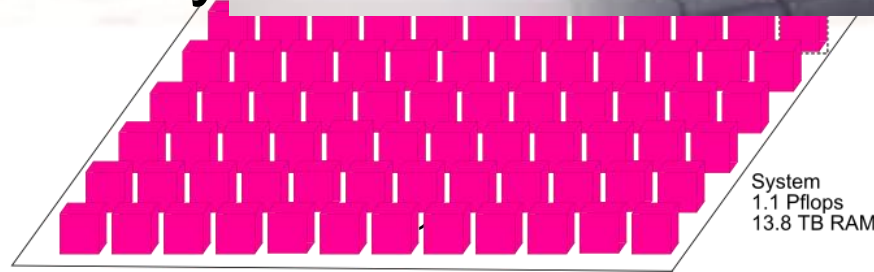
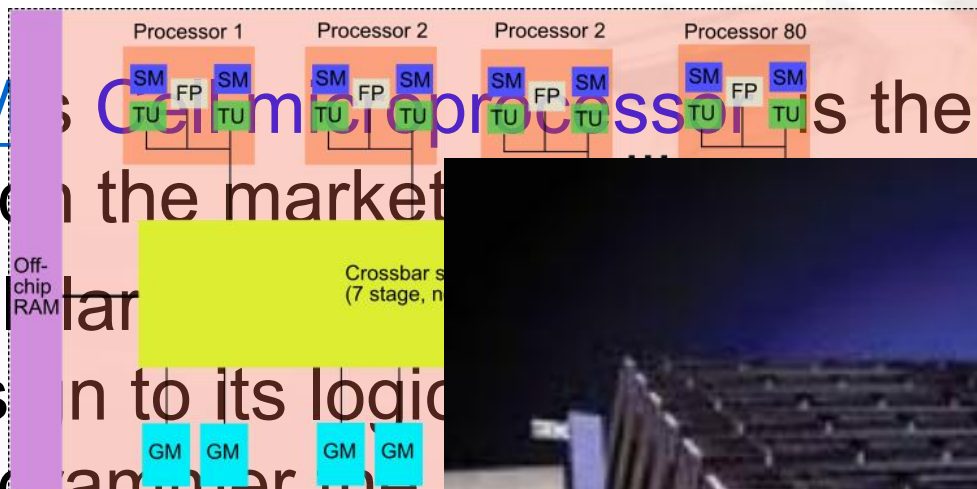
Cyclops64 (known as Blue Gene/C)

□ IBM's Cell microprocessor is the first one to

reach the market

□ Cellar

design to its logic
programmer the
concurrent threads
Each 'cell' is a co
units, memory, a
achieved by expl
inherent in many



System
1.1 Pflops
13.8 TB RAM



浙江大学
ZHEJIANG UNIVERSITY



Classes of Parallelism and Parallel Architecture

❑ Two kinds of parallelism in application

- Data-level Parallelism (DLP)
- Task-level Parallelism (TLP)

❑ Hardware exploit in 4 major ways

- Instruction level Parallelism
- Vector Architecture and GPUs
- Thread-Level Parallelism
- Request-Level parallelism



Classes of computers

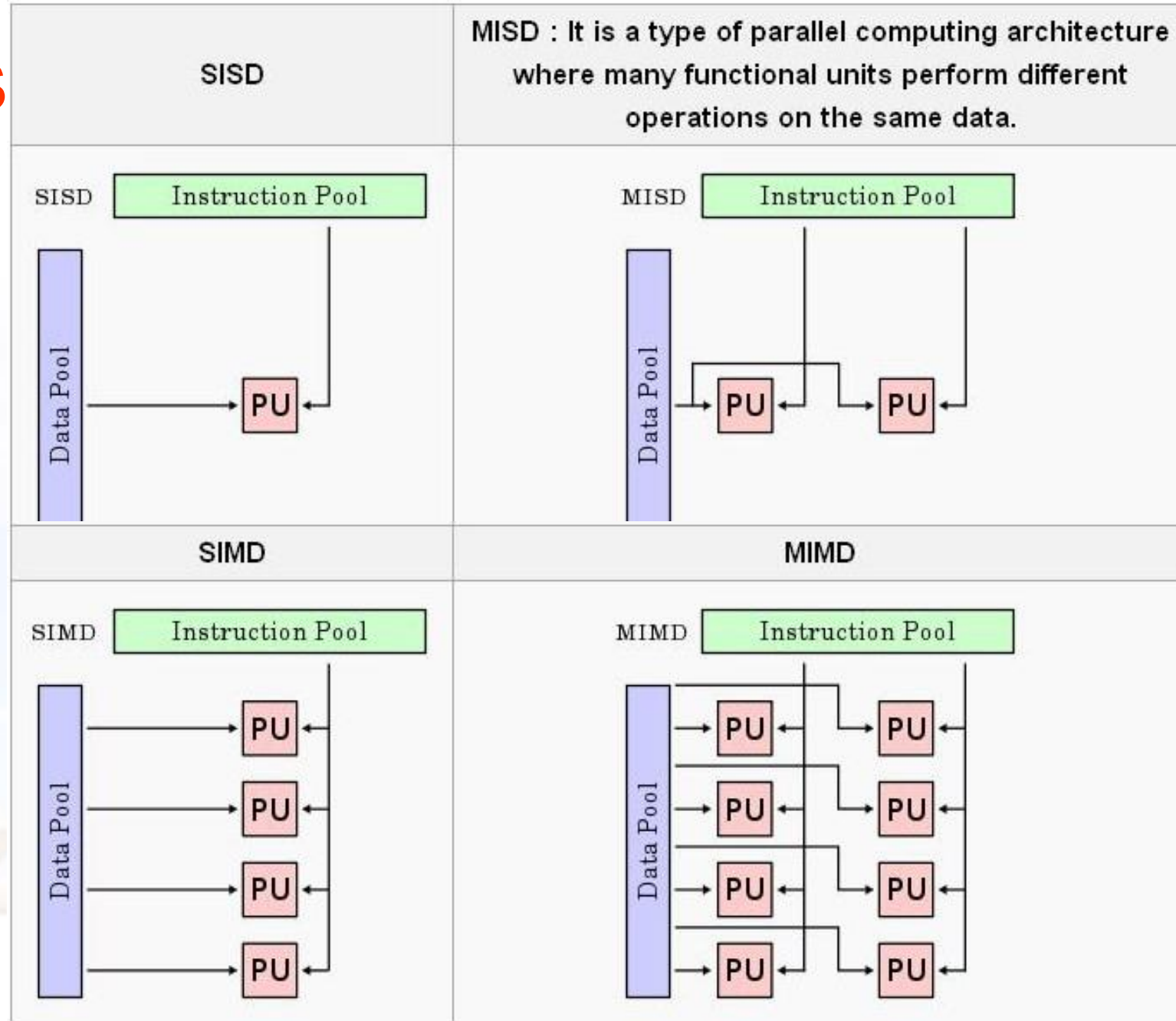
• **Flynn's Taxonomy:** A classification of computer architectures based on the number of streams of instructions and data



- **SISD (Single Instruction Single Data)**
 - Uniprocessors
- **MISD (Multiple Instruction Single Data)**
 - ???
- **SIMD (Single Instruction Multiple Data)**
 - Examples: Illiac-IV, CM-2
 - » Simple programming model
 - » Low overhead
 - » Flexibility
 - » All custom
- **MIMD (Multiple Instruction Multiple Data)**
 - Examples: SPARCCenter, T3D
 - » Flexible
 - » *Use off-the-shelf micros*



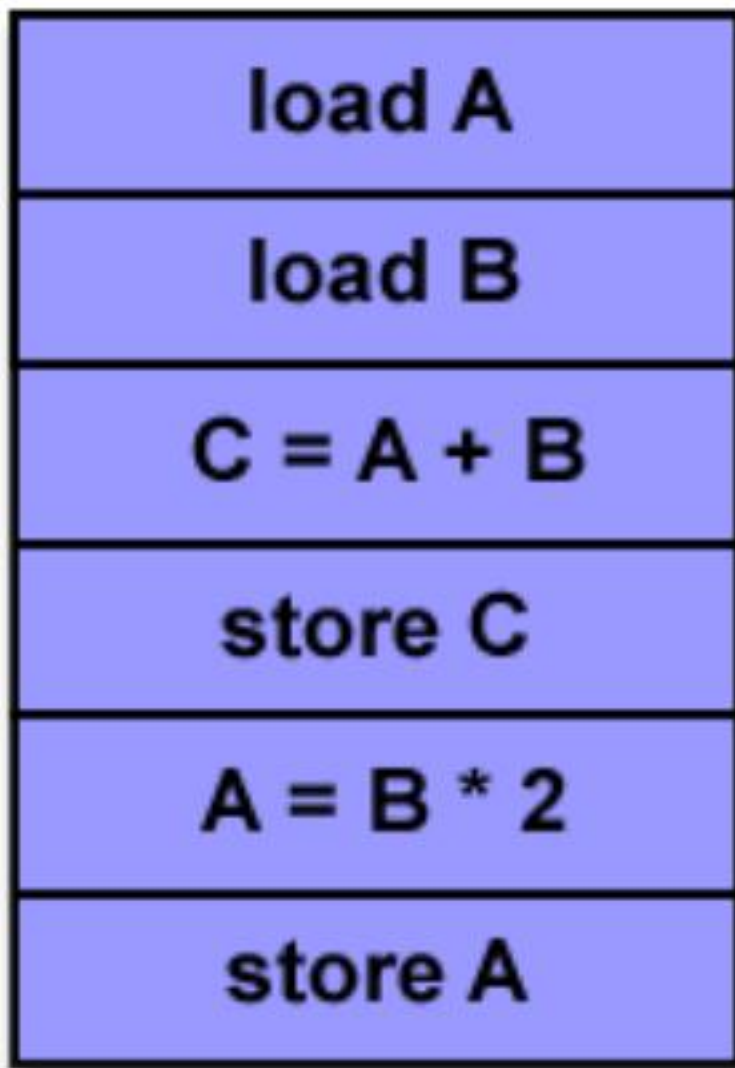
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SISD

- ❑ A serial (non-parallel)
- ❑ Single instruction being acted on each cycle
- ❑ Single data: one input during a cycle
- ❑ Deterministic
- ❑ This is the oldest prevalent form
- ❑ Examples: microcomputers and mainframes



stream is
the clock

being used as

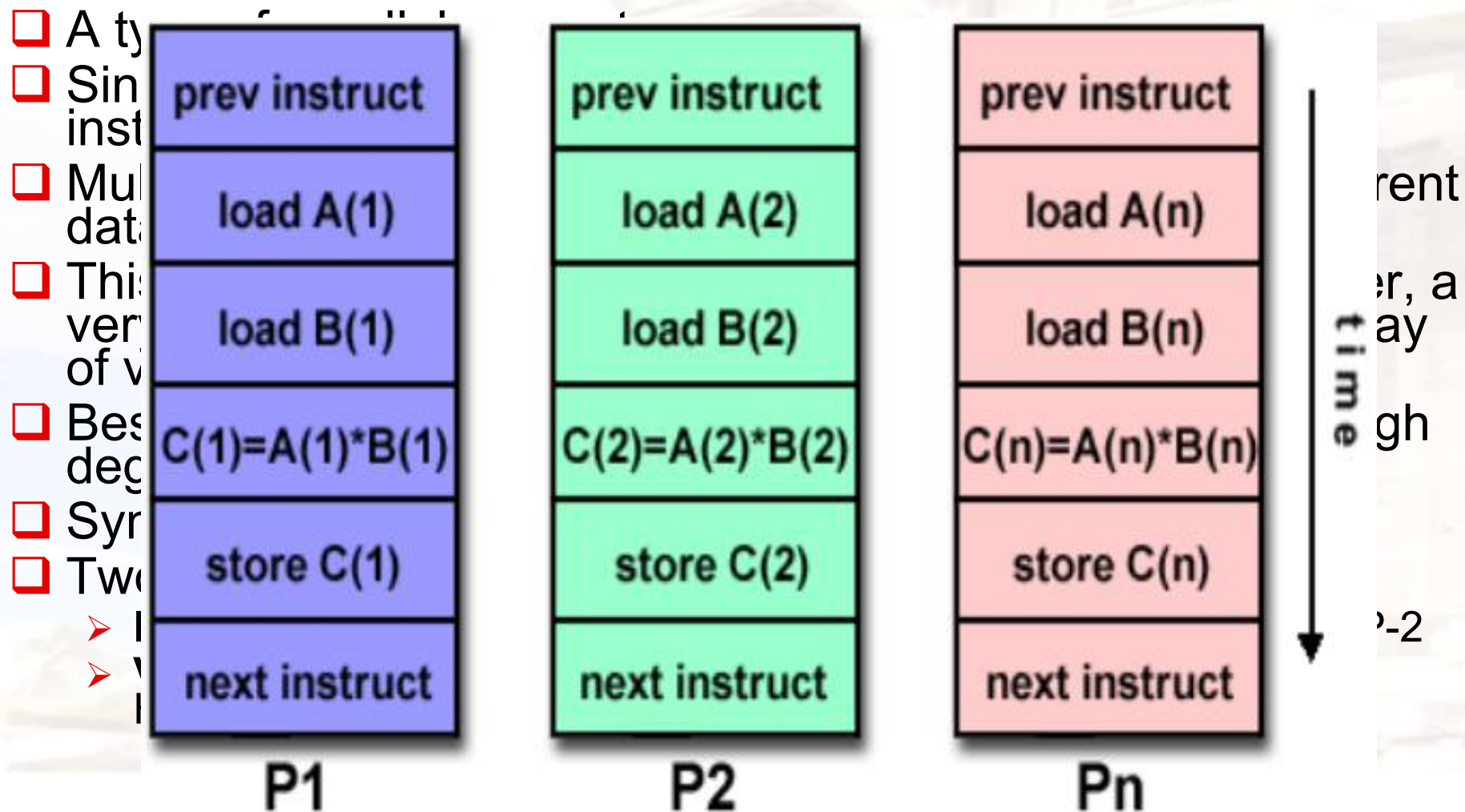
most

operations



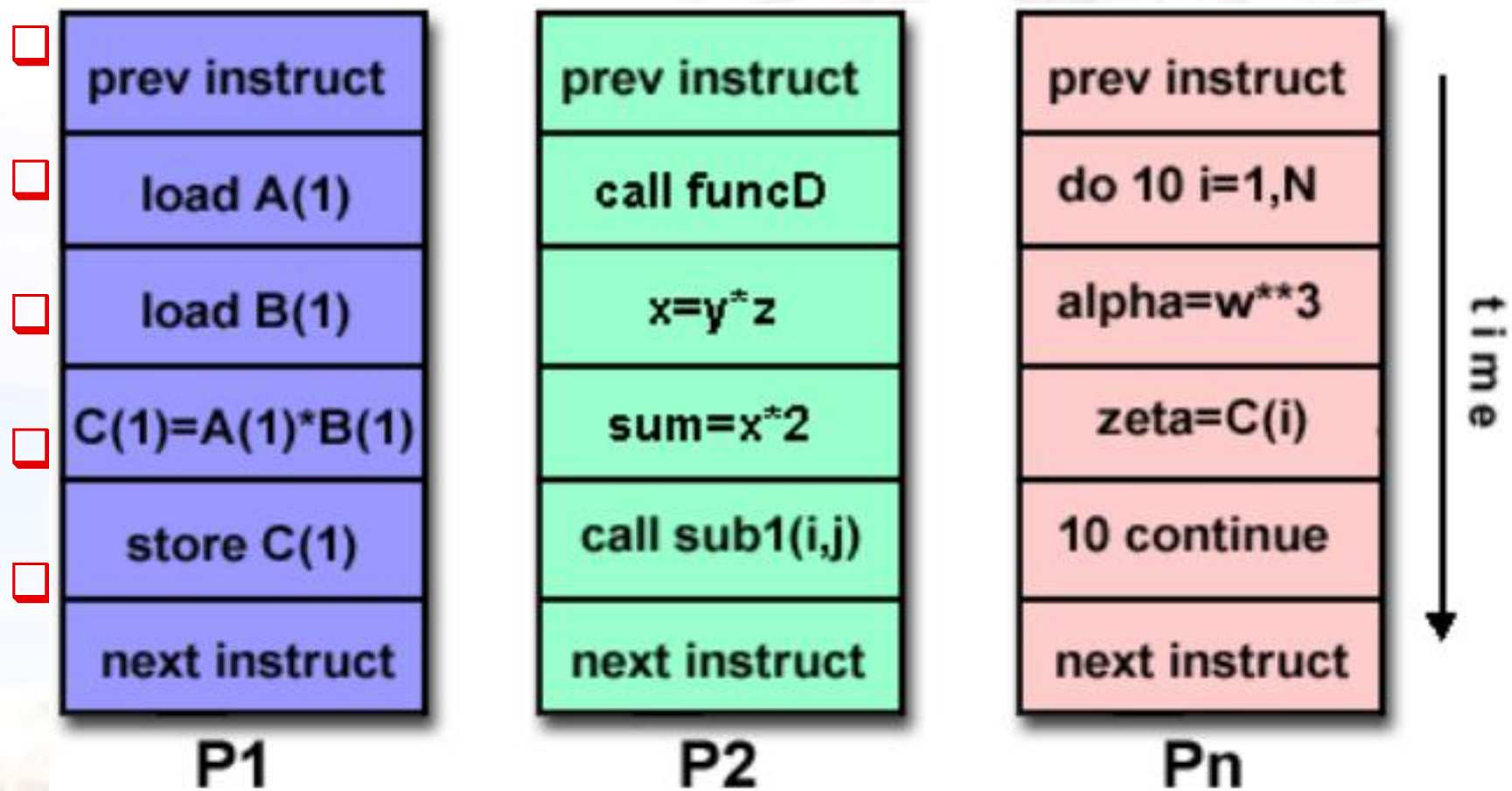


SIMD





MIMD





Topics in Chapter

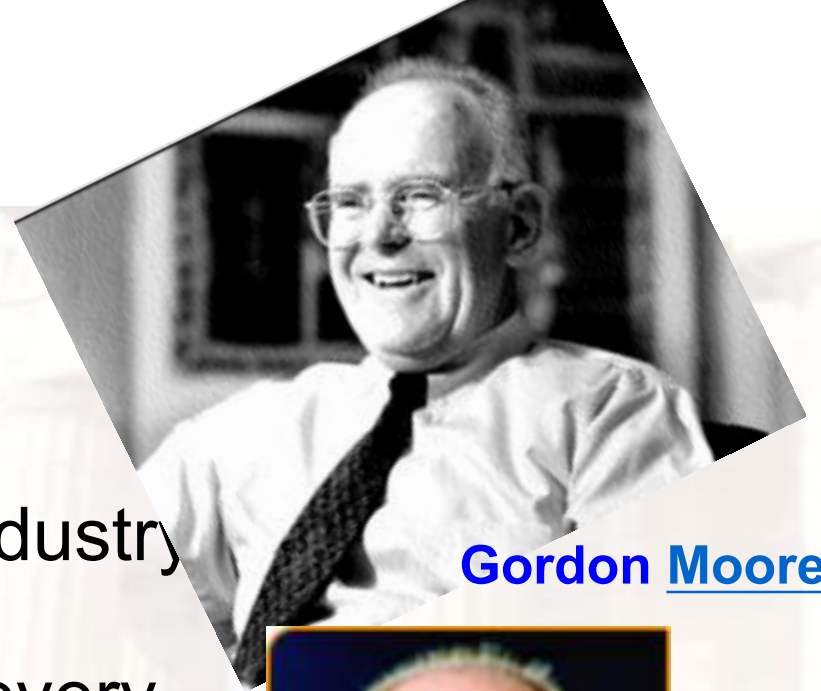
- 1.1 Why take this course ?
- 1.2 Classes of computers in current computer market
- 1.3 Defining computer architecture and What's the task of computer design?
- 1.4 Trends in Technology
- 1.5 Trends in power in Integrated circuits
- 1.6 Trends in Cost
- 1.7 Dependability
- 1.8 Measuring, Reporting and summarizing Perf.
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Technology Trends

❑ Moore Law

- ❑ In 1965 he predicted that the number of components the industry would be able to place on a computer chip would double every year. In 1975, he updated his prediction to once every two years. It has become the guiding principle for the semiconductor industry to deliver ever-more-powerful chips while decreasing the cost of electronics.



Gordon Moore





Gordon Moore on Moore's law

- ❑ **Moore's brief Bio**

<http://www.intel.com/pressroom/kits/bios/moore.htm>

- ❑ **Gordon Moore on Moore's law**

<http://www.sichinamag.com/Article/html/2007-09/2007919032802.htm>

- ❑ **Video** on conversation with Moore

<http://you.video.sina.com.cn/b/7076856-1282136212.html>





Technology Trends

❑ Integrated circuit logic technology

- Transistor(晶体管) Density: incr. 35% per year, (4x every 4 years)
- Die size: 10%-20% per year
- Transistor count per chip: 40-55% per year or 2x 10 to 24 months

❑ Semiconductor DRAM

- Capacity per DRAM: 25%-40% per year (2x every 2-3 years)
- Memory speed: about 10% per year
- Might stop and may be replaced

❑ Semiconductor Flash - standard storage device in PMDs

- Capacity per flash chip: 50-60% per year (2x every 2 years)
- 15-20 times cheaper than DRAM

❑ Magnetic Disk tech.

- Density: 30% p.y. Before 1990; 60% p.y. 1990-1996
- 100 p.y. 1996-2004; 30% p.y. after 2004
- capacity: about 60% per year

❑ Network

- bandwidth: 10Mb $\xrightarrow{10 \text{ years}}$ 100Mb $\xrightarrow{5 \text{ years}}$ 1Gb

Designers often design for the next technology.





Improvement of DRAM rate

AQA Edition	Year	DRAM growth	Characterization of impact on DRAM capacity
1	1990	60% / year	Quadrupling every 3 years
2	1996	60% / year	Quadrupling every 3 years
3	2003	40%-60%/ year	Quadrupling every 3 to 4 years
4	2007	40% / year	Doubling every 2 years
5	2011	25% - 40% / year	Doubling every 2 - 3 years



Import Notes

❑ A rule of thumb

- Cost decrease rate $\sim$ density increase rate

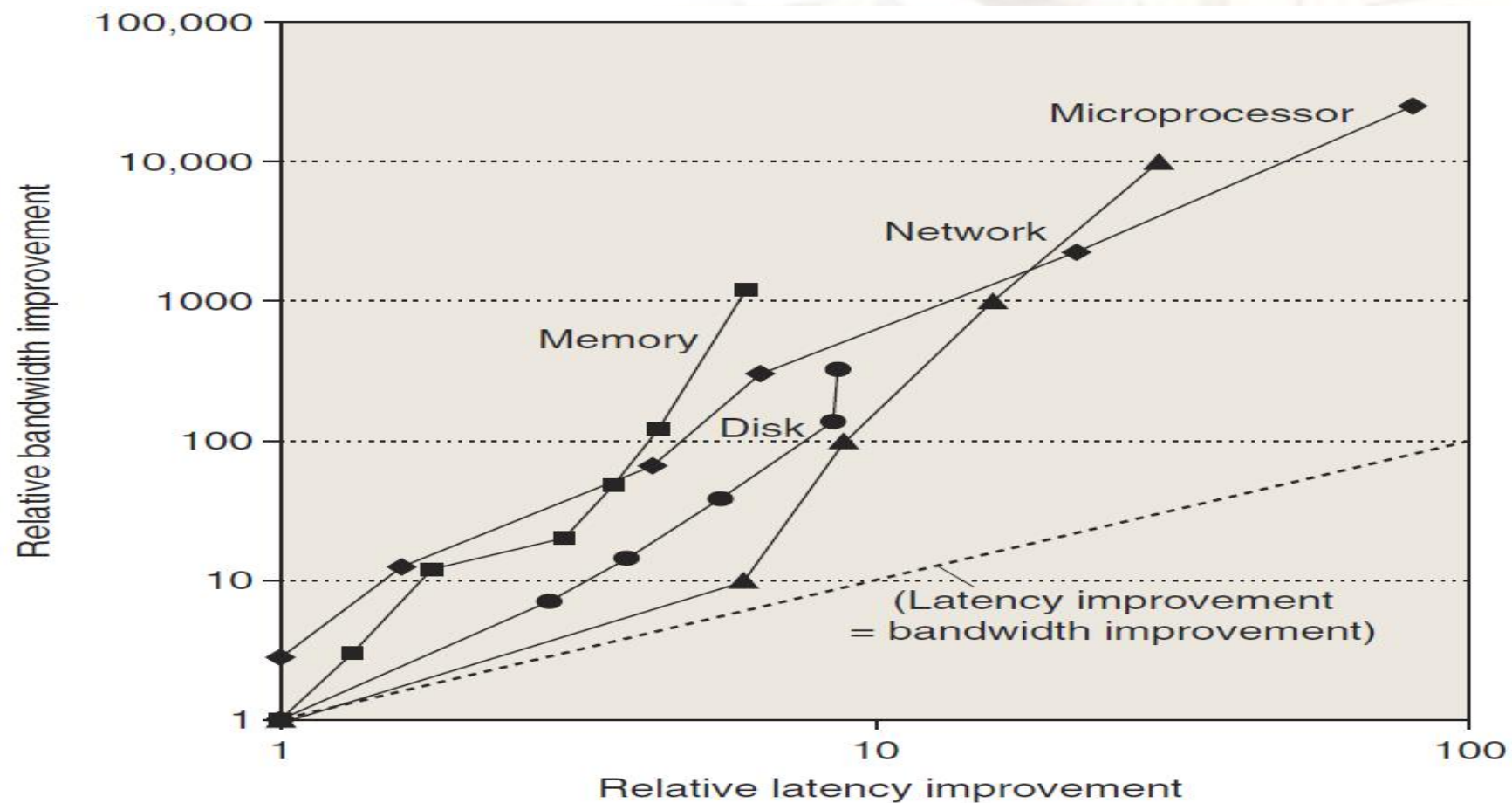
❑ Technology thresholds

- Technology improves continuously, an impact of this improvements can be in discrete leaps.





Perf. Trends: Bandwidth over latency



□ 10, 000X—25, 000X in Bandwidth
30X—80X in latency



Performance milestones in microprocessor

Microprocessor	16-bit address/bus, microcoded	32-bit address/bus, microcoded	5-stage pipeline, on-chip I & D caches, FPU	2-way superscalar, 64-bit bus	Out-of-order 3-way superscalar	Out-of-order superpipelined, on-chip 1.2 cache
Product	Intel 80286	Intel 80386	Intel 80486	Intel Pentium	Intel Pentium Pro	Intel Pentium 4
Year	1982	1985	1989	1993	1997	2001
Die size (mm ²)	47	43	81	90	308	217
Transistors	134,000	275,000	1,200,000	3,100,000	5,500,000	42,000,000
Pins	68	132	168	273	387	423
Latency (clocks)	6	5	5	5	10	22
Bus width (bits)	16	32	32	64	64	64
Clock rate (MHz)	12.5	16	25	66	200	1500
Bandwidth (MIPS)	2	6	25	132	600	4500
Latency (ns)	320	313	200	76	50	15



Challenges for IC Technology

- ❑ IC characteristic: **feature size**(特征尺寸)
 - 10 microns in 1971 → 0.18microns in 2001
 - → 0.09 microns in 2006 → 0.032 microns in 2011
 - 7nm is under way.
 - **Rule of thumb**: transistor perf. Improves linearly with decreasing feature size.
- ❑ IC density improvement is both opportunity and Challenge:
 - **signal delay** for a wire increase in proportion to the product of its resistance and capacitance.
- ❑ **Wire delay**---major design limitation



□ “There is no reason anyone would want a computer in their home.”

--Ken Olson, president,
Chairman and founder of Digital Equipment Corp., 1977